git

VERSION CONTROL

What is git?

'sion Control System

- Software that allows you to track changes in a file system:

Traditional sion Control System

User Interface Parallel	2017/09/30 11:40 ...	LabVIEW Instrume...	178 KB
User Interface	2016/09/06 07:16 ...	LabVIEW Instrume...	231 KB
User Interface2	2016/09/06 10:33 ...	LabVIEW Instrume...	234 KB
User Interface3.1	2016/09/28 08:27 ...	LabVIEW Instrume...	239 KB
User Interface3	2016/09/15 08:22 ...	LabVIEW Instrume...	237 KB
User Interface4	2016/10/03 10:03 ...	LabVIEW Instrume...	248 KB
User Interface5	2016/10/11 10:08 ...	LabVIEW Instrume...	433 KB
User Interface6	2017/01/24 10:56 ...	LabVIEW Instrume...	379 KB
User Interface7	2017/04/11 02:09 ...	LabVIEW Instrume...	275 KB
User Interface8.2	2017/04/11 02:56 ...	LabVIEW Instrume...	247 KB
User Interface8	2017/04/11 10:47 ...	LabVIEW Instrume...	254 KB
User Interface9	2017/04/19 12:56 ...	LabVIEW Instrume...	194 KB
User Interface10.aliases	2017/10/19 01:09 ...	ALIASES File	1 KB
User Interface10.lvlps	2017/08/11 06:14 ...	LVLPS File	1 KB
User Interface10	2017/08/11 06:14 ...	LabVIEW Project	14 KB
User Interface10	2017/10/20 04:56 ...	LabVIEW Instrume...	199 KB
User Interface11	2017/09/25 11:42 ...	LabVIEW Instrume...	167 KB
User Interface12.aliases	2017/12/24 01:49 ...	ALIASES File	1 KB
User Interface12.lvlps	2017/10/31 12:03 ...	LVLPS File	1 KB
User Interface12	2017/10/31 12:03 ...	LabVIEW Project	17 KB
User Interface12	2017/10/31 11:51 ...	LabVIEW Instrume...	203 KB
User Interface13	2017/10/12 04:58 ...	LabVIEW Instrume...	187 KB

- HAL_Test
- Integ_HAL_R3
- Integration_HAL_R2
- Integration_HAL_R3
- Integration_HAL_R3_FIFO
- Integration_HAL_R4
- Integration_HAL_R5
- Integration_HAL_R6
- Integration_R1
- Integration_R2
- Integration_R2_FIFO
- Integration_R2_HAL
- Integration_R2_TIP

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- Coordinates work on the file system among multiple people.

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Distributed Version Control System

- Software that allows you to track changes in a file system:
  - Effectively - Changes are actually tracked.
  - Efficiently - Tracking is done intelligently.
- Coordinates work on the file system among multiple people.
  - Each team member has a local copy of the entire file system and history which is also stored remotely.
  - Individual changes can be merged after review.

Advantages of git?

- Complete file change history allows roll-back to previous versions when bugs break software.
- Team members can work concurrently on the same file system by creating a *branch* for each user. These can be *merged* once individual developments are complete.
    - A branch can be used to trial changes without altering the main source code of a project.
- Changes are traceable and are annotated to detail their rationale - Greatly improves the readability of code and aids with understanding why code is written in the way it is.

The git workflow

fetch / edit / stage / commit

1. Fetch the latest version of the files from the remote server.
2. Edit files in workspace.
3. Add files to the Staging Area (or Index).
4. Commit these staged files to the project history.

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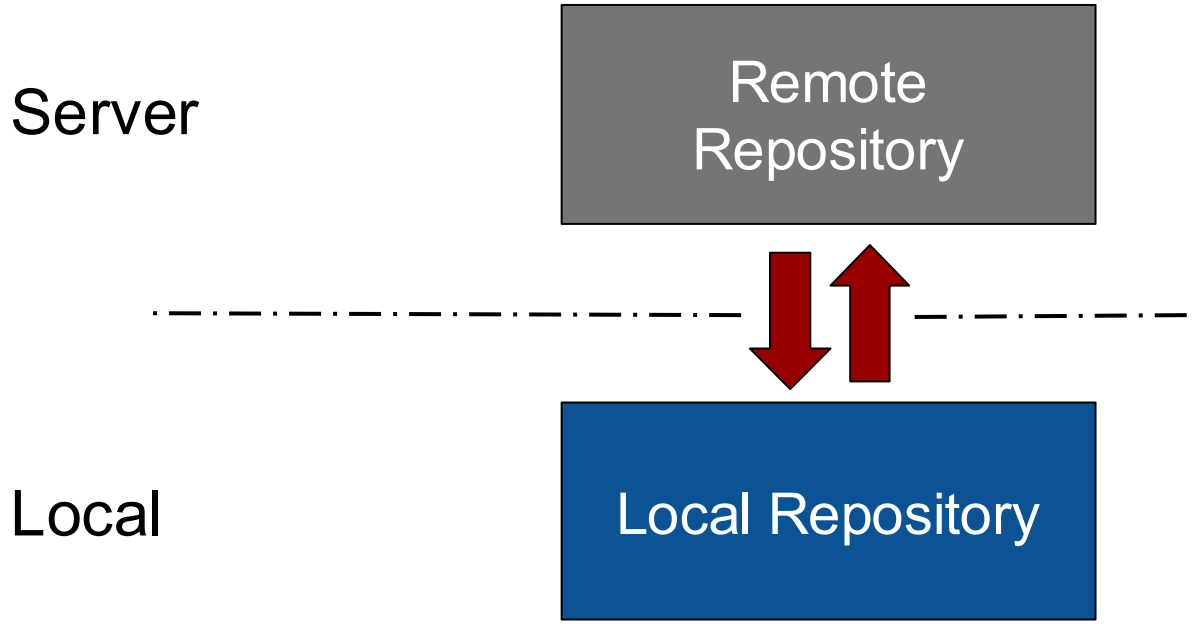
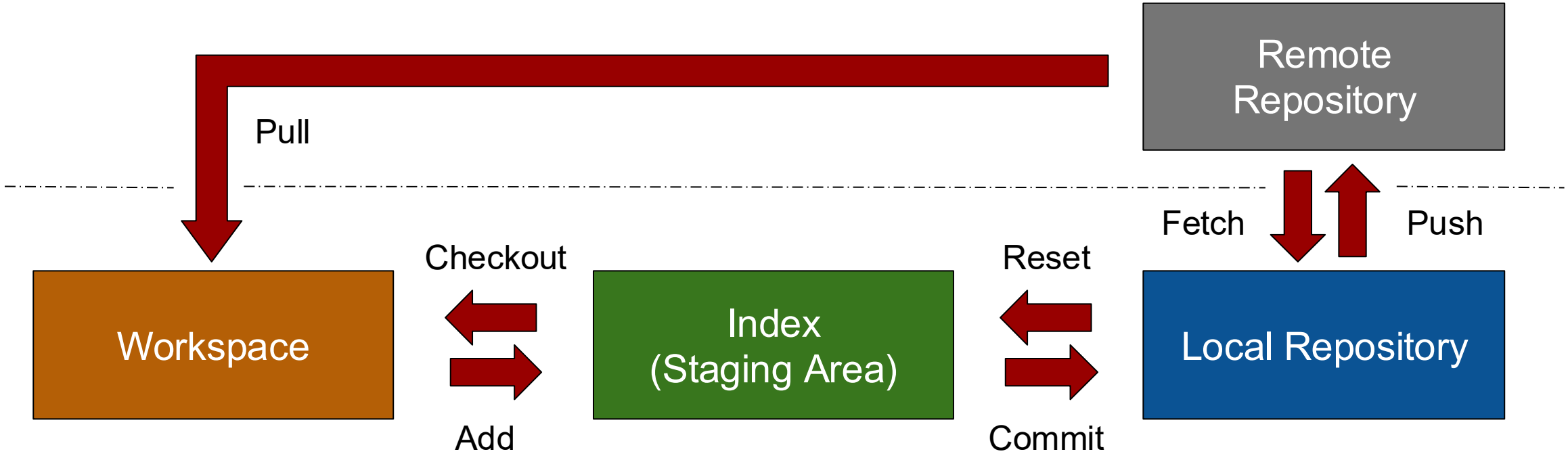
Where are these stored?

Repositories

- Store the source code and commit history of the file system.

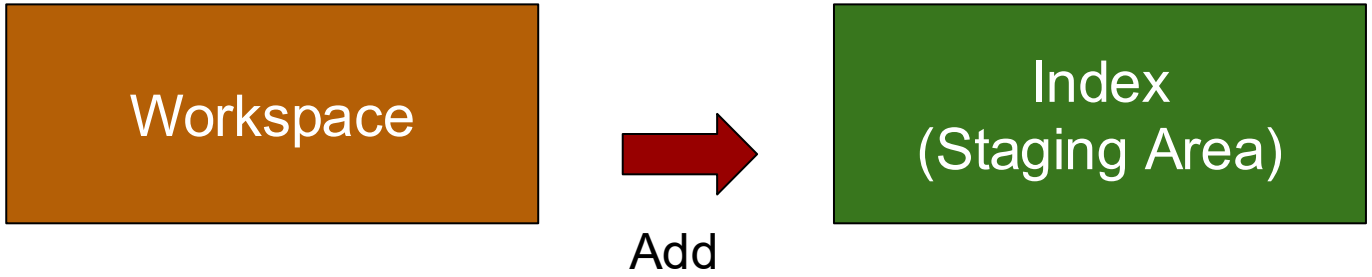


The git workflow



git Commands - Add

- Adds a file from the Workspace to the Index
- The Index is a staging area where files can be marked for update at the next commit.



git Commands - Commit

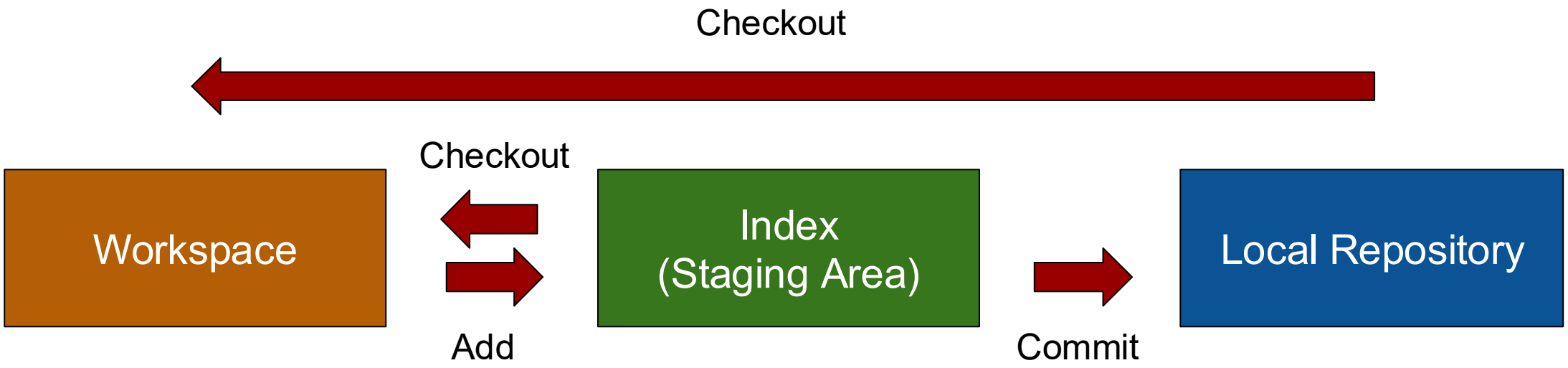
- Commits the staged files to the project history.
- Files are “safe” once committed and won’t be changed unless you specifically request a change.

git Commands - Checkout

- Allows you to view a previously staged file.

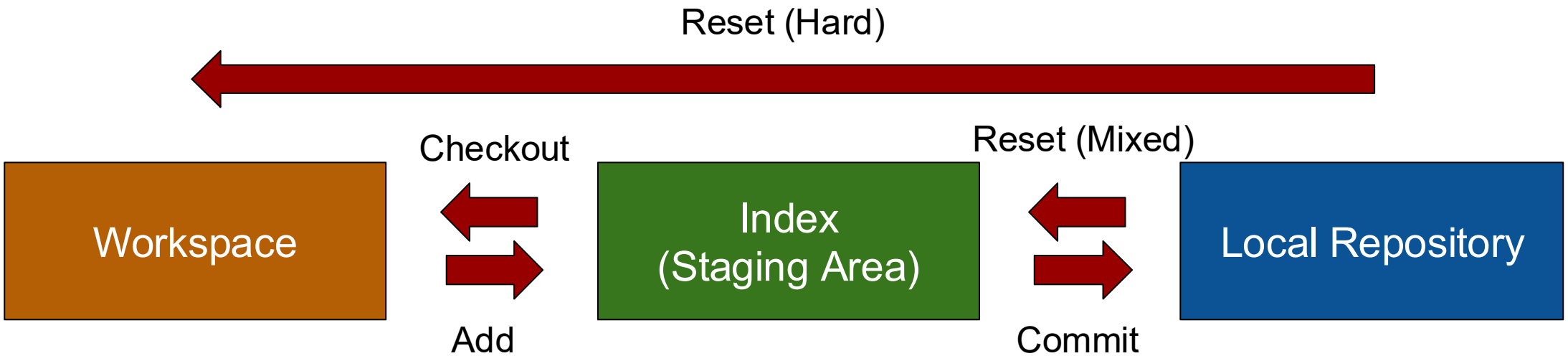
git Commands - Checkout

- Allows you to view a previously staged file.
- OR previous commit.



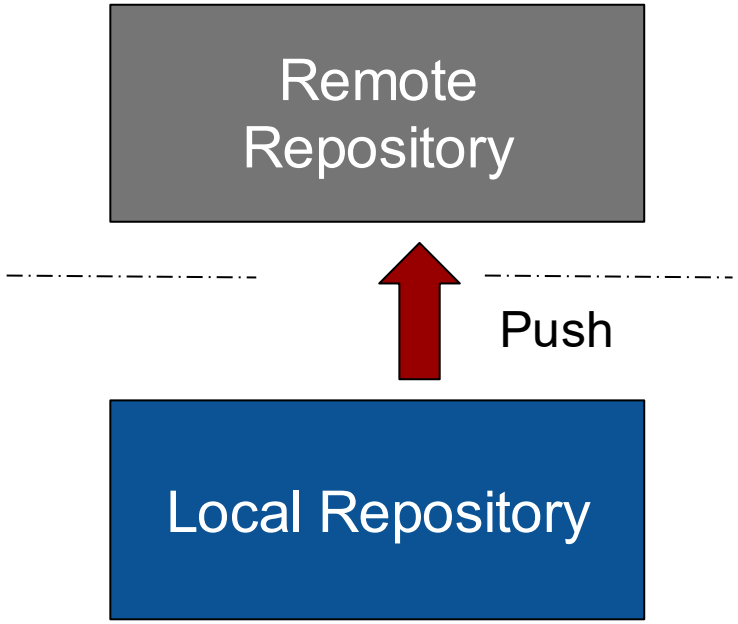
git Commands - Reset

- Allow you to update the index to a previous commit.
- OR allows you to update workspace as well



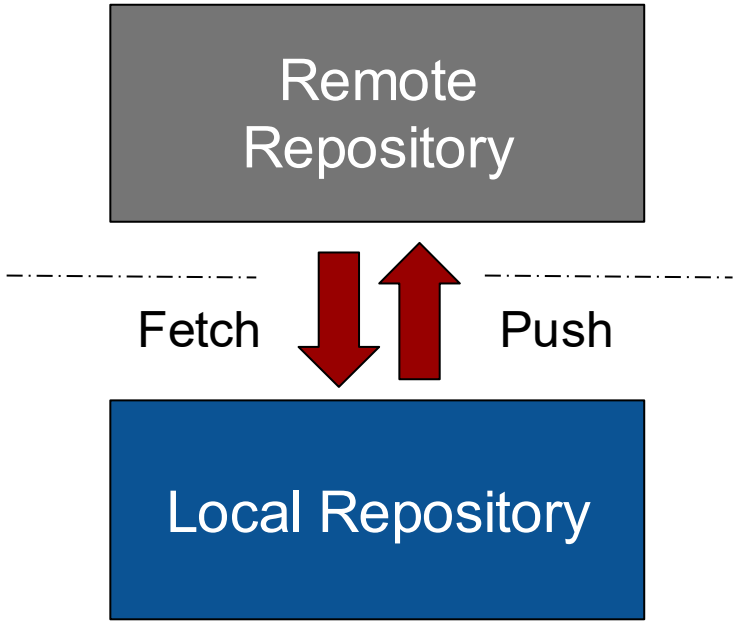
git Commands - Push

- Updates your remote repository with the state of your local repository.

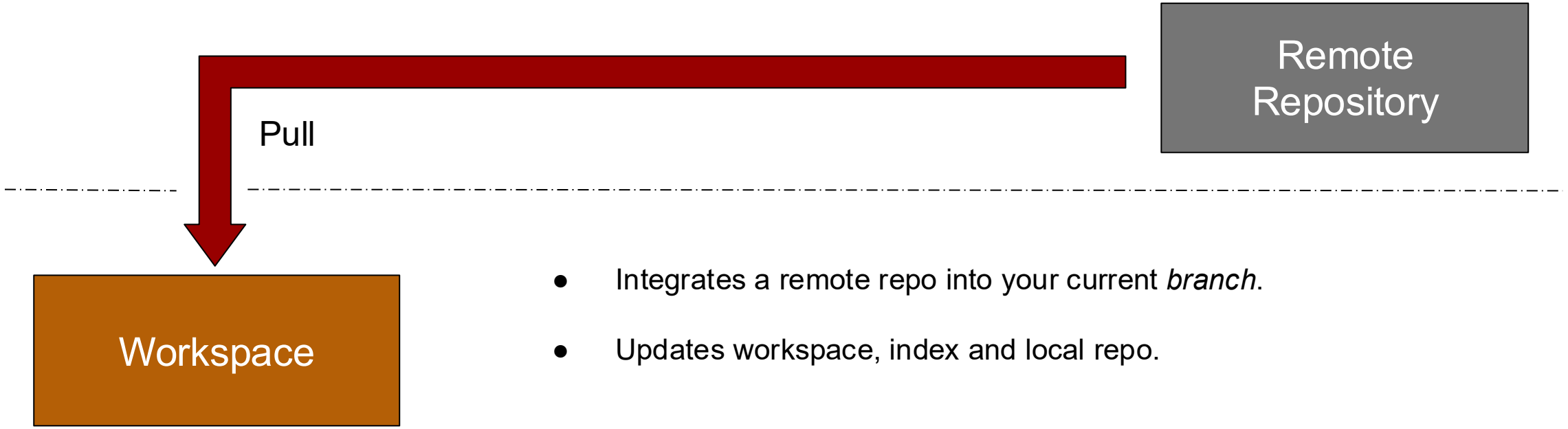


git Commands - Fetch

- Updates your local repository with the state of your remote repository.

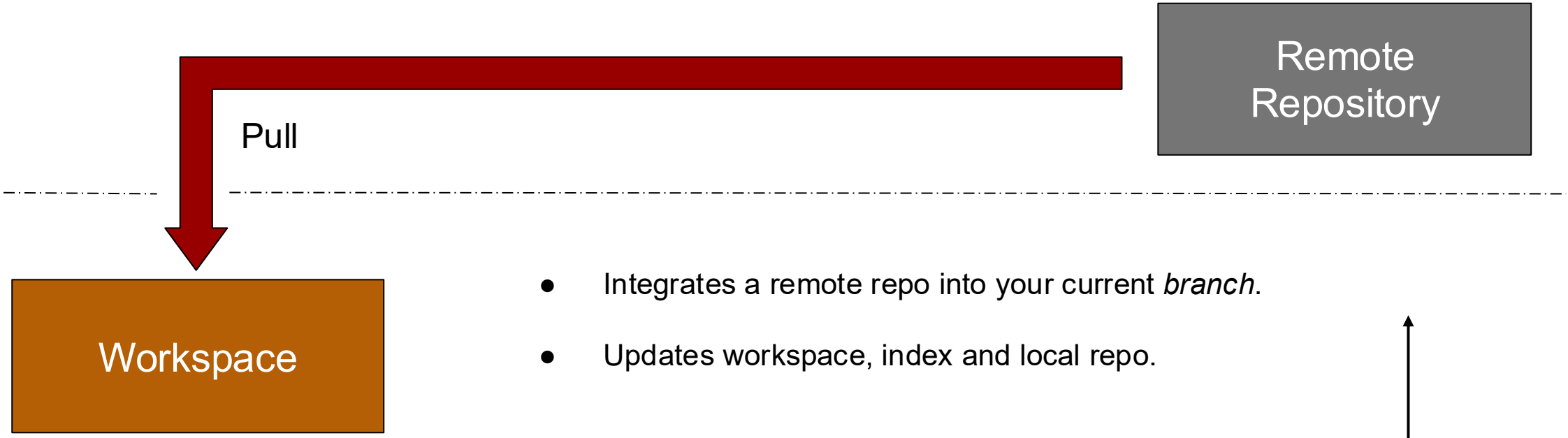


git Commands - Pull



- Integrates a remote repo into your current *branch*.
- Updates workspace, index and local repo.

git Commands - Pull

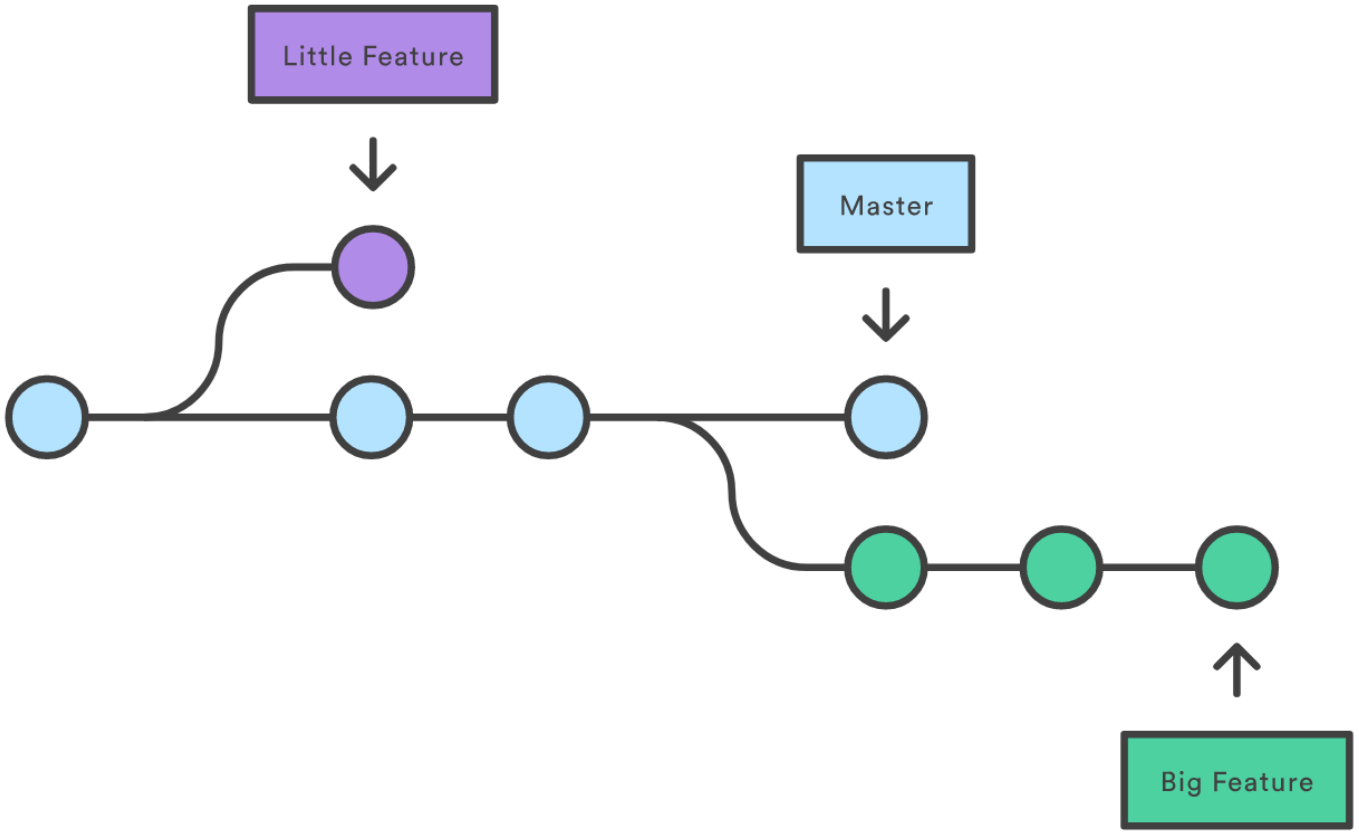


- Integrates a remote repo into your current *branch*.
- Updates workspace, index and local repo.

What is a branch?

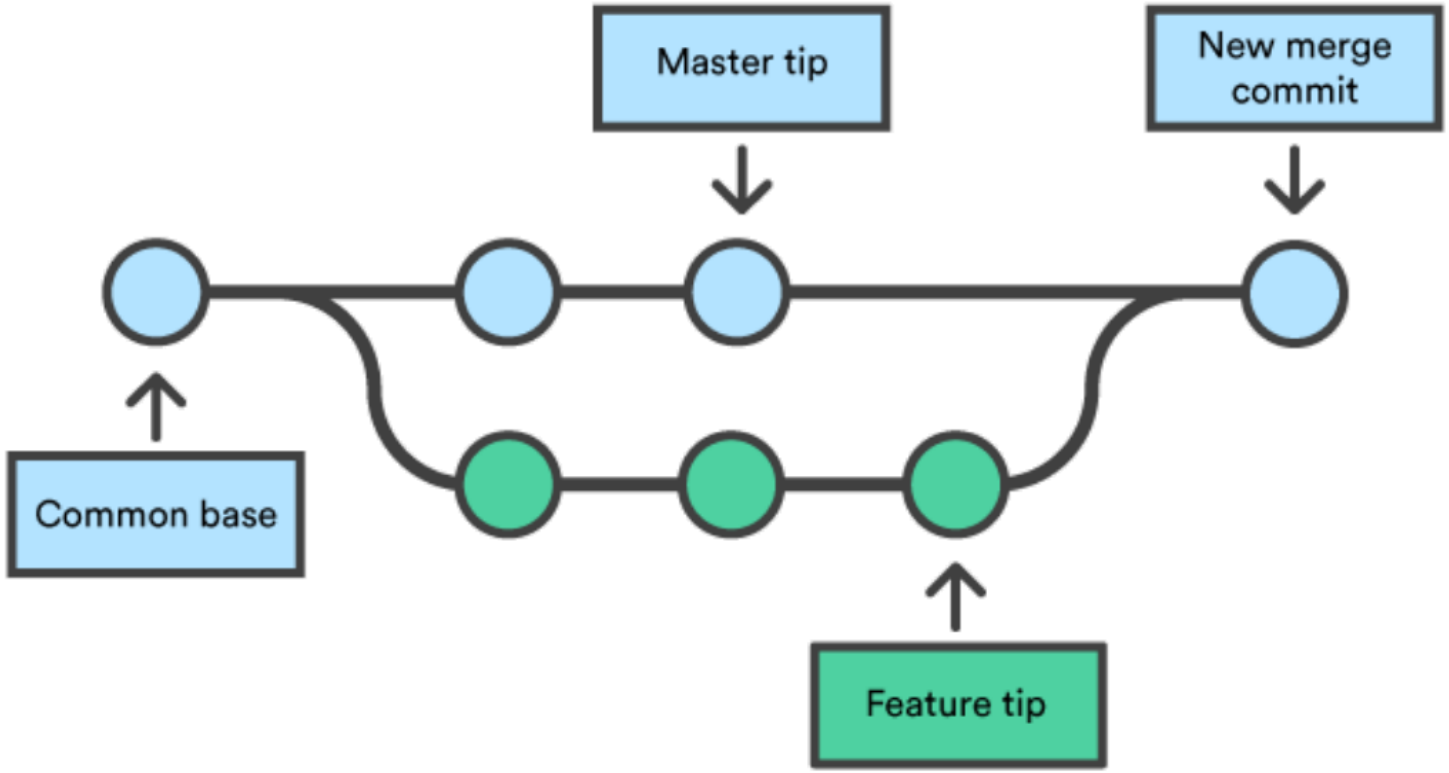
git Commands - Branch

- Allows isolated lines of development for a repository.
- Multiple users can work on separate tasks independently without modifying the master code.
- A single user can create a branch to test new functions/bug fixes without changing the master.



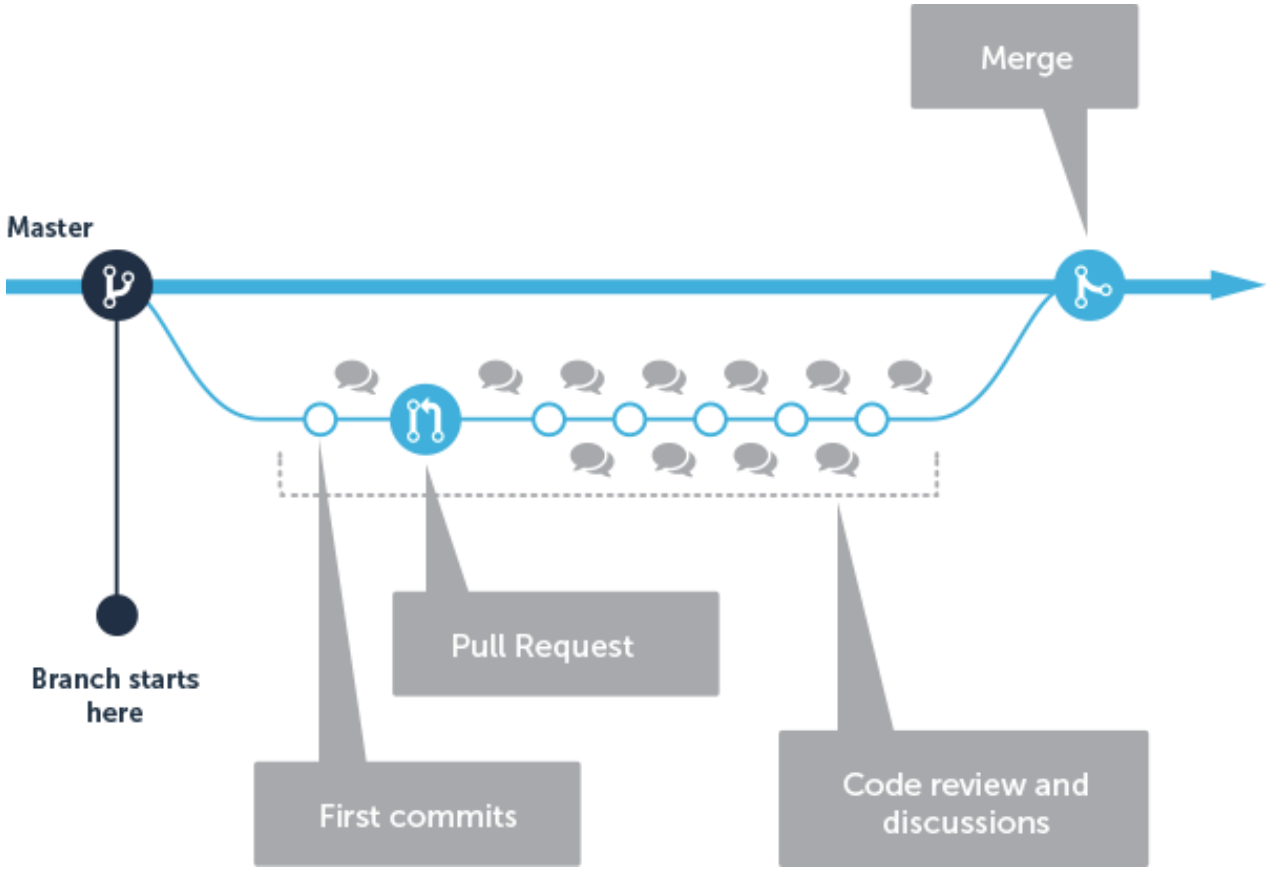
git Commands - Merge

- Branches can be merged together if development on both has been approved.
- A merge updates the currently selected branch and its history.
- Automatic combining of history.
- If errors occur these must be resolved before merge can complete.



git Commands - Pull Request

- Provides a user-friendly web interface for discussing proposed changes before integrating them into the official project.



References

Images from:

- https://www.eclipse.org/artwork/
- https://www.gitcolony.com/features
- https://www.atlassian.com/git/tutorial
- https://bitbucket.org/

External resources:

- https://docs.github.com/en/get-started/start-your-journey/git-and-github-learning-resources
- https://www.atlassian.com/git/tutorials
- https://git-scm.com/docs/gittutorial